STAFF REVIEW · H446 026 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. A stack removes the most recently added item (LIFO); a queue removes the earliest added item (FIFO).

2. The values must be sorted in the order expected by the search.

3. 18. C# arrays start at index 0.

4. Indexes 3 → 5 → 4 (values 16 → 31 → 23). The search narrows the sorted range each time.

I Do — source worked steps

1. What switches ON? — CODE:
Q = door unlocks — Start with the result the question is asking us to control.

2. What are the alternatives? — CODE:
A ∨ B — A card OR PIN works. OR (∨) means at least one; both can also be true. Do not write ⊻ here: XOR would wrongly stop access when both card and PIN are valid.

3. What must also be true? — CODE:
(A ∨ B) ∧ C — The building must be open , so join the options with AND.

4. What must be OFF? — CODE:
¬D — Lockdown must not be active. NOT reverses D.

5. Join the chunks — CODE:
Q = (A ∨ B) ∧ C ∧ ¬D — Work backwards: the final AND checks all requirements. Draw OR, AND and NOT first, then connect them to the final AND.

6. Use intermediate columns — For Q = (A ∨ B) ∧ ¬C, three inputs give $2^3 = 8$ combinations. Write every input row, calculate A∨B, then ¬C, then combine those two columns with AND. — CODE:
A B C | A∨B | ¬C | Q
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 1 | 1 | 1
1 1 1 | 1 | 0 | 0 — Example row 010: A∨B = 1 and ¬C = 1; therefore Q = 1. Always complete the intermediate columns before Q.

We Do — guided CHECK

1. AND: both required
CODE:

Q = A ∧ B
Output: login. No OR options or NOT exception. AND joins two requirements.
CODE:
A B | Q
0 0 | 0
0 1 | 0
1 0 | 0
1 1 | 1

2. OR or XOR?
Choose the gate and write Q = A ⊻ B for exactly one . How would the symbol and truth table change for at least one ?
CODE:
A B | EXACTLY ONE | AT LEAST ONE
0 0 | ? | ?
0 1 | ? | ?
1 0 | ? | ?
1 1 | ? | ?
XOR (⊻) versus OR (∨)
Exactly one → XOR: Q = A ⊻ B . At least one → OR: Q = A ∨ B (both allowed).
CODE:
A B | XOR | OR
0 0 | 0 | 0
0 1 | 1 | 1
1 0 | 1 | 1
1 1 | 0 | 1

3. Build two chunks
Write the options in brackets. Apply NOT to the exception . What final gate must join them?
CODE:
Q = (A ? B) ? ?C
Sketch the circuit before revealing the answer.
OR, NOT, then AND
CODE:
Q = (A ∨ B) ∧ ¬C
Two options → OR. Jam absent → NOT C. Both conditions → final AND.

4. Words → gates
Name the output Q.
Bracket the alternative triggers.
Join the power requirement.
Invert maintenance lock.
Write the full expression, then draw.
Can you identify the final gate by working backwards?
Compare each chunk
CODE:
Options (A ∨ B)
Requirement C
Exception ¬D

CODE:
Q = (A ∨ B) ∧ C ∧ ¬D
Draw OR(A,B), AND with C, NOT D, then a final AND . Only reveal once pupils have sketched the circuit.

You Do — source mark schemes

1. CODE:
A = door sensor; B = motion sensor
C = alarm armed; D = test mode

Q = (A ∨ B) ∧ C ∧ ¬D

1. OR A and B.
2. NOT D.
3. AND (A∨B) with C.
4. AND that result with ¬D to give Q.
Check: A=1, B=0, C=1, D=0 gives Q=1. Changing D to 1 makes Q=0. A three-input final AND is also correct.

2. P = ¬(A ∨ B) ⊻ C. Calculate OR, then NOT, then XOR. XOR is 1 when its two inputs differ.
CODE:
A B C | A∨B | ¬(A∨B) | P
0 0 0 | 0 | 1 | 1
0 0 1 | 0 | 1 | 0
0 1 0 | 1 | 0 | 0
0 1 1 | 1 | 0 | 1
1 0 0 | 1 | 0 | 0
1 0 1 | 1 | 0 | 1
1 1 0 | 1 | 0 | 0
1 1 1 | 1 | 0 | 1
Check: For 000, NOT(0 OR 0)=1 and 1 XOR 0 = 1.

3. T = temperature high; L = load high; M = maintenance mode on.
CODE:
F = (T ∨ L) ∧ ¬M

T L M | T∨L | ¬M | F
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 1 | 1 | 1
1 1 1 | 1 | 0 | 0
Check: At least one trigger must be high, and maintenance must be off.

4. CODE:
Q = (A ⊻ B) ∧ ¬C

1. XOR A and B.
2. NOT C.
3. AND those outputs.

A B C | A ⊻ B | ¬C | Q
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 0 | 1 | 0
1 1 1 | 0 | 0 | 0
Check: Q=1 only when A and B differ and C=0.

Exit Ticket — source answer

1. True. Each of the three Boolean inputs has two states, giving 2 × 2 × 2 = 8 combinations.